# ECE435: Medical Imaging Project 2

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## Introduction

This project develops and implements a processing pipeline for spectral domain optical coherence tomography (SD-OCT). Provided with raw data captured by a CCD line camera, this pipeline transforms spectral-domain measurements into interpretable spatial-domain A-scan, B-scan, and M-scan outputs. By analyzing samples such as layered Scotch tape and a vibrating speaker, this project highlights the impact of signal processing steps to improve image quality and recover structural information.

## Part I: Guiding Computations

Given the following parameter values:

$$\begin{aligned} \text{Center Wavelength } \lambda_0 &= 1310 \text{ nm} \\ \text{Axial Pixel Size } dz &= 3.6\ \mu\text{m} \\ \text{Bandwidth } \Delta\lambda &= 100 \text{ nm} \\ \text{NA } &= 0.055 \\ \text{n } &= 1 \end{aligned}$$

The lateral resolution is given by the formula and plugging values results in:

$$\delta x = \delta y \approx 0.37 \frac{\lambda_0}{NA} = 8.746\ \mu\text{m}$$

The lateral resolution is limited by the numerical aperture and center wavelength of the imaging system. We want finer resolution so we would want higher numerical aperture (which results in more detailed images) and a smaller center wavelength. The tradeoff is that this lowers our penetration depth.

The axial resolution is given by the formula and plugging values results in:

$$\delta z \approx \frac{2 \log 2}{n\pi} \frac{\lambda_0^2}{\Delta\lambda} = 7.458\ \mu\text{m}$$

The axial resolution is limited by the source bandwidth. For a given center wavelength, having a larger bandwidth decreases the axial resolution $\delta z$, which corresponds to finer axial resolution.

To find the B-scan pixel aspect ratios, begin with the size of the B-scan file. Loaded into MATLAB using `fread` as type `uint16`, the B-scan is of size 20838400. Thus dividing by 2048 entries per scan, $M + D = 10175$. Given $D = 175$ backgrounds, the B-scan consists of $M = 10000$ A-scans. We are given the axial pixel size, $dz = 3.6\mu$m, and since the A-scan consists of 2048 pixels, the total axial depth is 7.373mm. Accounting for the mirror imaging artifact of OCT, the actual axial depth should be $7.373/2 = 3.686$mm.

To find the lateral pixel size, divide the given the width of B-Scan (1mm) by the total amount of A-scans. Thus, the lateral pixel size is $dz = 1\text{mm}/10000 = 0.1\ \mu$m. Therefore our B-scan pixel aspect ratio is

$$\frac{dz}{dx} = 36$$

meaning the pixel is 36 times as long (z-direction) as it is wide (x-y direction). The overall B-scan aspect ratio is 1mm $\times$ 3.686 mm accounting for the mirror image artifact and 1mm $\times$ 7.373 mm without.

## Part II: The A-Scan function

To generate a complex A-Scan in the spatial domain, the function requires the background scans, the actual A-scans, and the L2K matrix. All processing begins in the lambda domain which includes subtracting the average background measurement, deconvolving the signal by dividing it by a *third-order polynomial fit of the background measurement, and then applying a hamming window function to smooth the data edges. Then the processed data is converted to the k-domain via multiplication with the L2K matrix and finally, an inverse FFT transforms the data into a complex A-scan back in the spatial domain.

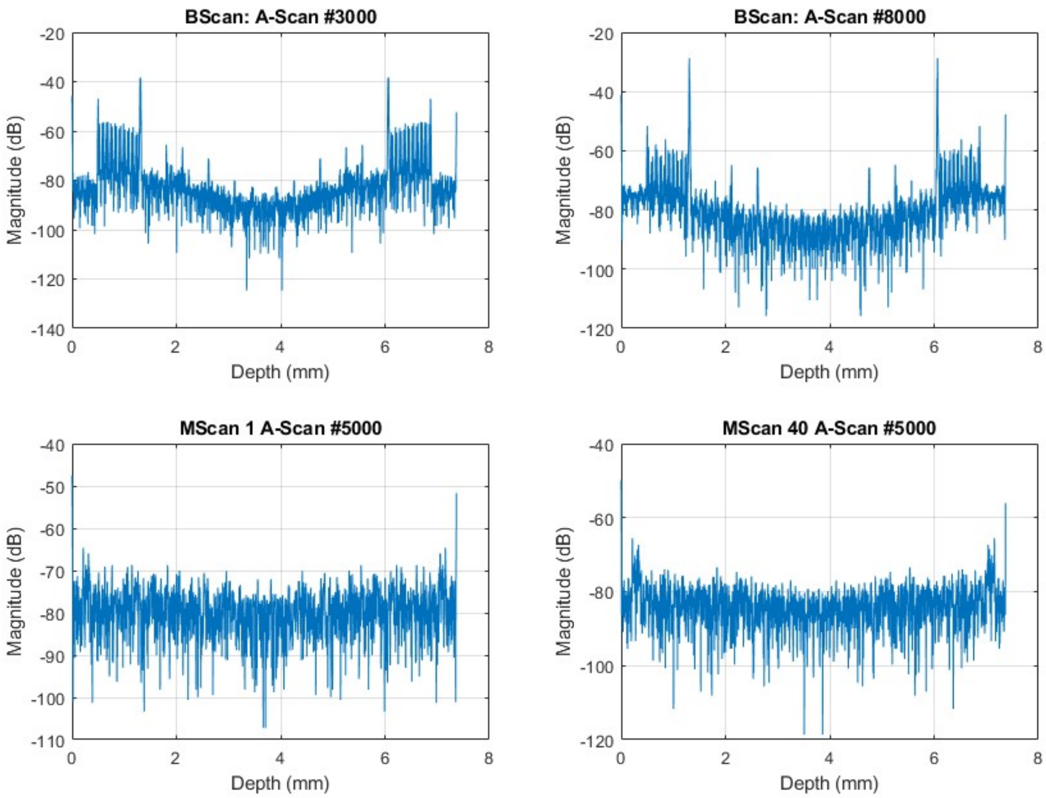


Figure 1: Comparison of A-Scan magnitudes taken from `BScan Layers.raw`, `MScan1`,and `MScan40`

Table 1: Breakdown of processing time (seconds) for each A-scan in Figure 1

| **Processing Step** | **B-Scan** | **M1** | **M40** |
|---|---|---|---|
| Background Subtraction | 0.0234s | 5.4182s | 0.2833s |
| Deconvolution | 0.0353s | 1.7232s | 0.3553s |
| Windowing | 0.0252s | 6.9560s | 1.4241s |
| Convert to k-domain (L2K) | 1.5071s | 21.2167s | 34.5913s |
| FFT Processing | 0.1022s | 12.7378s | 25.3320s |

The breakdown of the processing times for each step in the function can be seen in Table 1. Across all the scans, converting to the k-domain took the longest amounmt of processing time. This is expected as to convert, we multiplied with the $2048 \times 2048$ L2K matrix. The next longest step was taking the IFFT and converting back into the spatial domain. **A comment:** the first time I tried to run this code, my computer was unresponsive for 5 minutes with task manager reporting 100% CPU and RAM usage. To assuage this, I converted the data from MATLAB's native `double` to `single` and also clearing the opened .raw files after processing them.

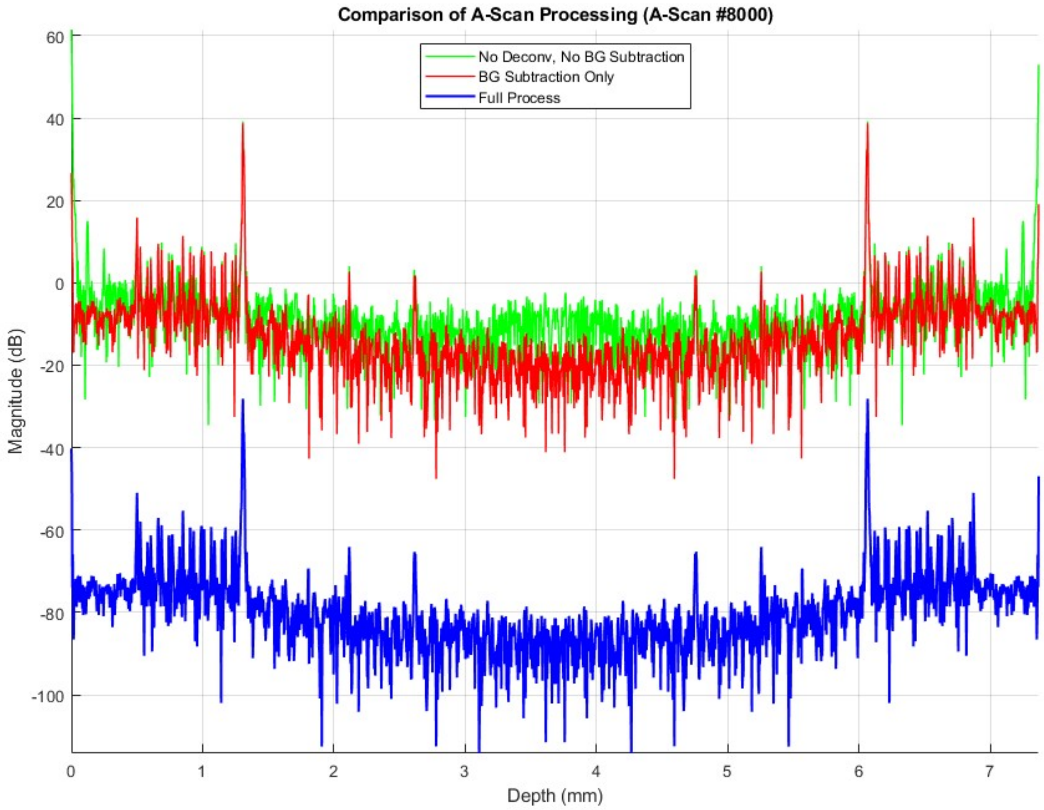


Figure 2: Comparison of A-scan magnitude of the 8000th index of `BScan Layers.raw` with no deconvolution, and no deconvolution with no background subtraction.

The comparison of the three A-scan processing methods highlights how each step cleans the signal. The unprocessed signal contains a massive peak at 0 mm (the DC term) and a high level of background noise from the system which can be seen by the fairly level signal. After applying background subtraction, it is evident that some noise has been removed, making the actual peaks of the signal much easier to distinguish. With the full processing, the signal has improved contrast and is now normalized (via the IFFT). This in turns shifts the magnitude down by 80 dB.

## Part III: The B-Scan

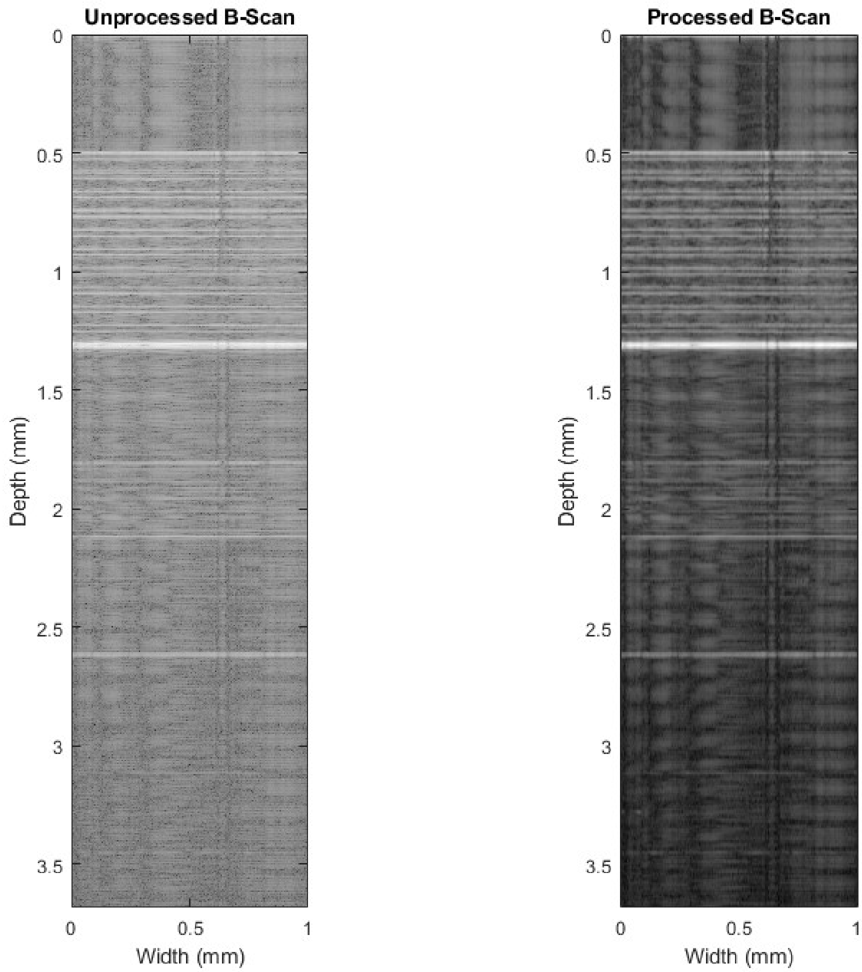


Figure 3: Comparison of Unprocessed and Processed B-Scan

Referring to section 1, the axial pixel size, $dz = 3.6\mu$m, the lateral pixel size is $dz = 1\text{mm}/10000 = 0.1\ \mu$m, and thus the B-Scan pixel aspect ratio $\frac{dz}{dx} = 36$. The overall B-scan aspect ratio is 1mm $\times 3.686$ mm accounting for the mirror image artifact and 1mm $\times 7.373$ mm without.

To process the image, the data was converted to a logarithmic scale and normalized so that the contrast between the bright surface and faint internal reflections could be seen. In the unprocessed B-Scan image, black speckles (high-frequency spatial noise) are visible and so to remedy that, a Gaussian filter with $\sigma = 1.2$ was applied to smooth out these higher frequencies. Observe in the processed B-Scan that these spurious black speckles no longer appear. This fairly simple processing pipeline effectively enhances contrast and suppresses speckle noise without overprocessing the image.

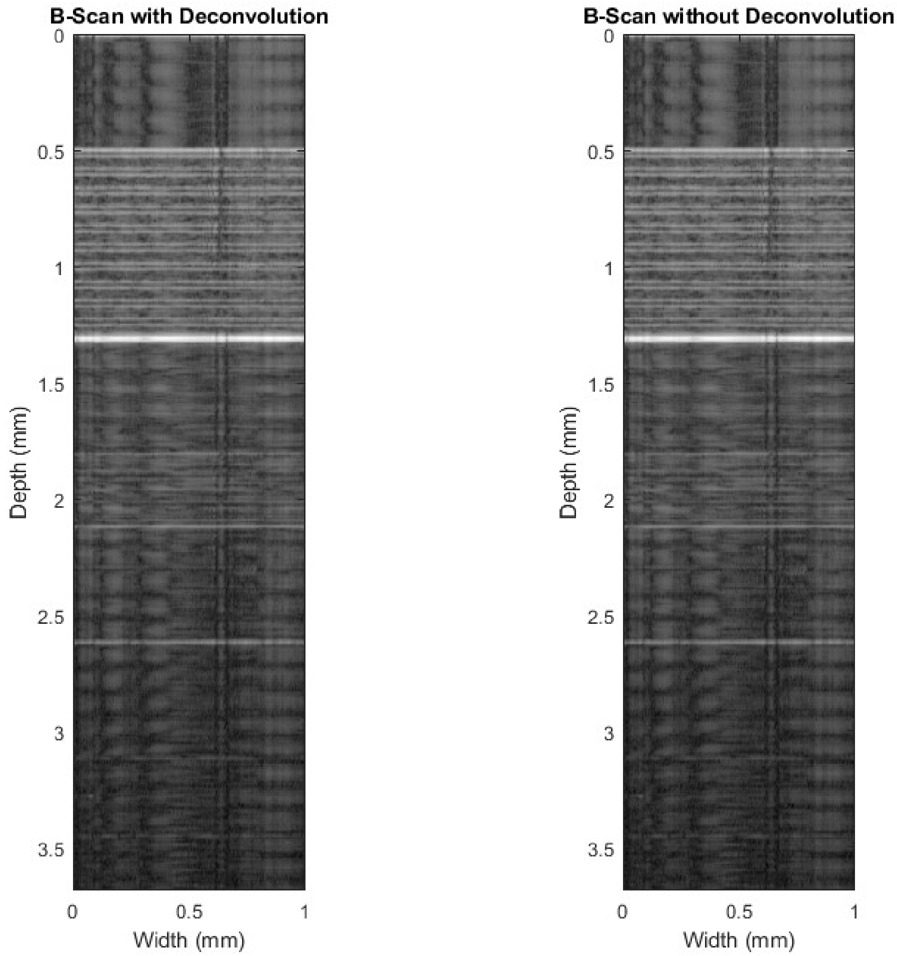


Figure 4: Comparison of Processed B-Scans with and without Deconvolution

Observing Figure 5, there does not seem to be much difference.This is expected since referring in our processing stage, we elected to normalize the peak intensity to 0 dB.

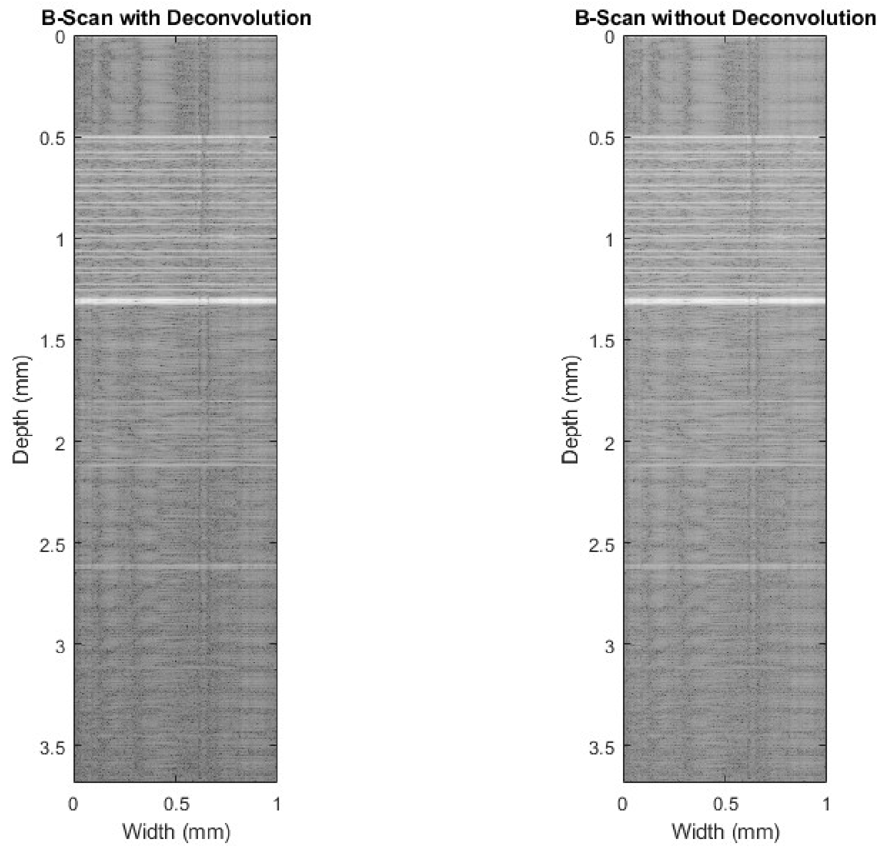


Figure 5: Comparison of Unprocessed B-Scans with and without Deconvolution

Removing the processing step, we see that the B-Scan with deconvolution improves image contrast and axial resolution.

Assuming the bright bands beginning around 0.5mm depth correspond to the surfaces of the tape, from Figure 5, there appears to be 10 layers of tape. The thin black bands in between the bright bands which correspond to the thickness of the tape measure (using fine tick marks in MATLAB) to be roughly 0.03mm. The larger gaps between the bright bands corresponding to distance between each layer measure to be roughly 0.05mm.

## Part IV: The M-Scan

The average of the A-Scan magnitudes from the 1-tone M-Scan is shown in Figure 6. Ignoring the peak at 0mm, the following two highest peaks at depth 0.216mm and depth0.3348mm correspond to the 61st and 94th pixels since $dz = 0.0036$ (and accounting for 1-indexing).

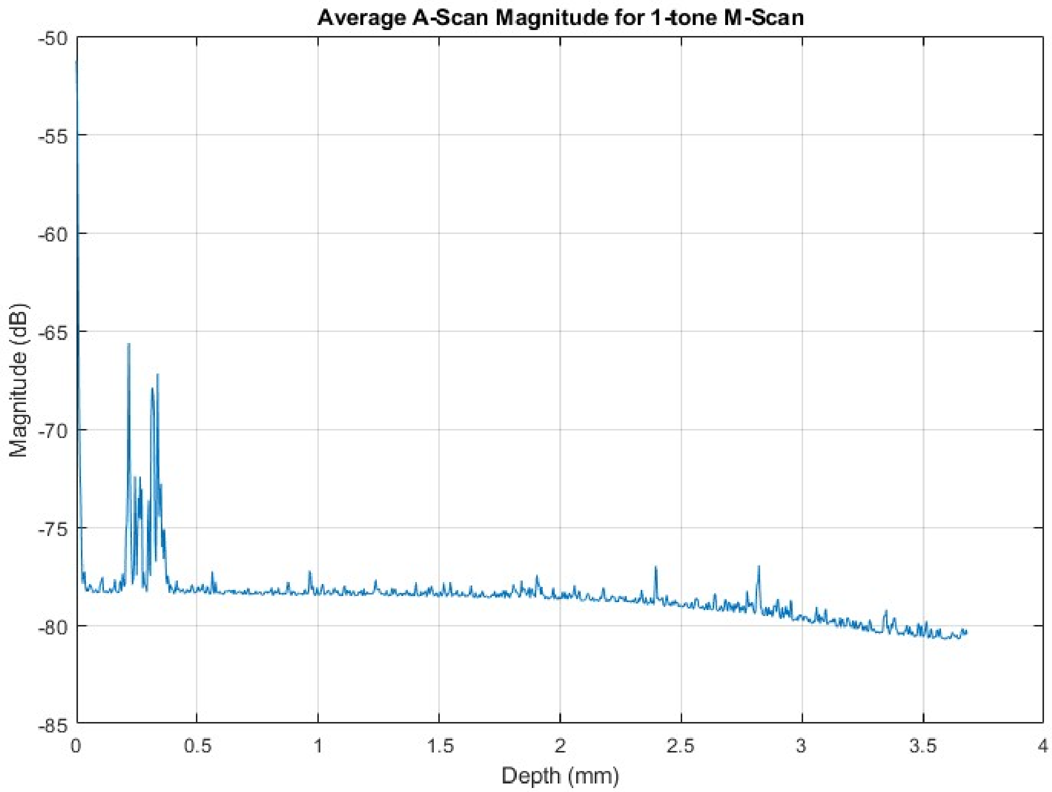


Figure 6: Average of the A-Scan magnitudes from the 1-tone M-Scan

To track sub-pixel motion, spectral domain phase microscopy was applied at these pixel locations, and the resulting displacement time series and frequency spectrum are shown.

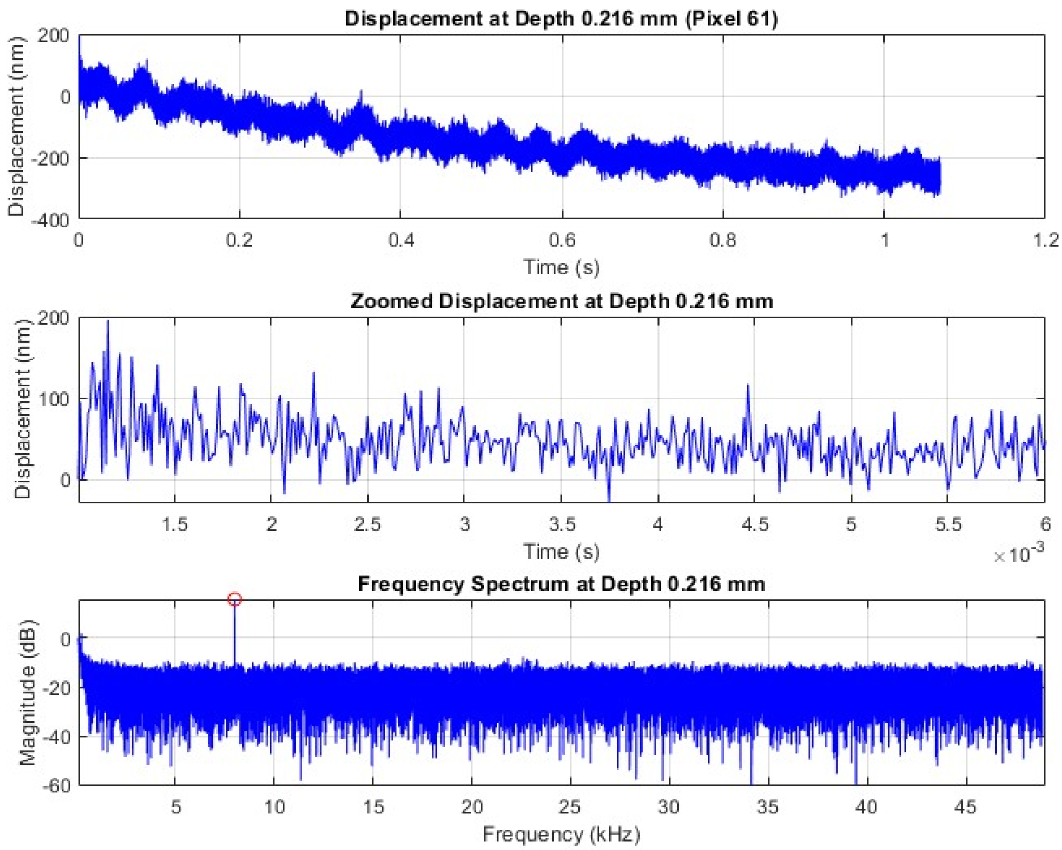


Figure 7: SDPM Time and Frequency Domain Data at Pixel 61 for 1-tone M-Scan

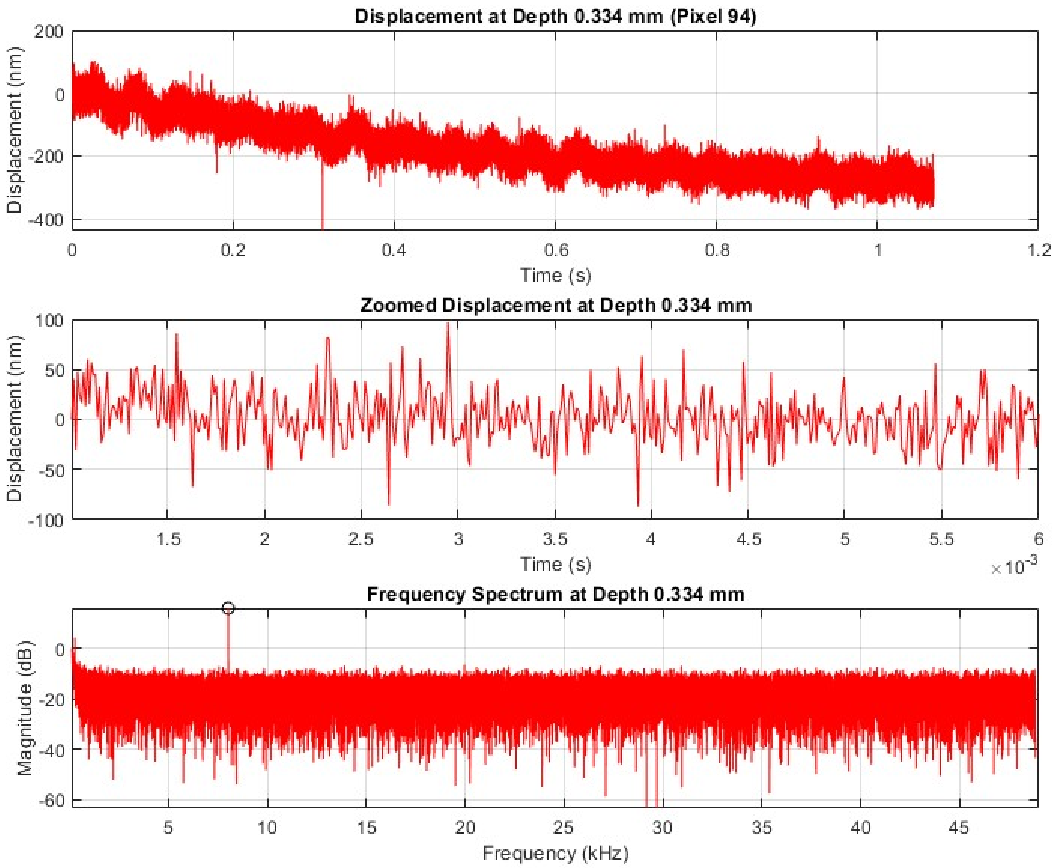


Figure 8: SDPM Time and Frequency Domain Data at Pixel 94 for 1-tone M-Scan

For both pixel 61 and pixel 94, both the time and frequency domain data look normal. Zooming in, we see a sinusoidal oscillation, indicating some periodic signal. Looking at the frequency spectrum, there is a distinct peak (circled in red and black) at 8.0109kHz for both pixels. Thus the tone played through the speaker was a 8.0109kHz tone.

These steps were repeated for the 40-tone M-Scan. The average of the A-Scan magnitudes from the 40-tone M-Scan is shown in Figure 9. Just like the 1-tone M-Scan, the pixels reflecting the peak magnitudes are the same. Applying the same analysis to pixel 61 and pixel 94 for the 40-tone M-Scan, we see the results in Figure 10 and Figure 11.

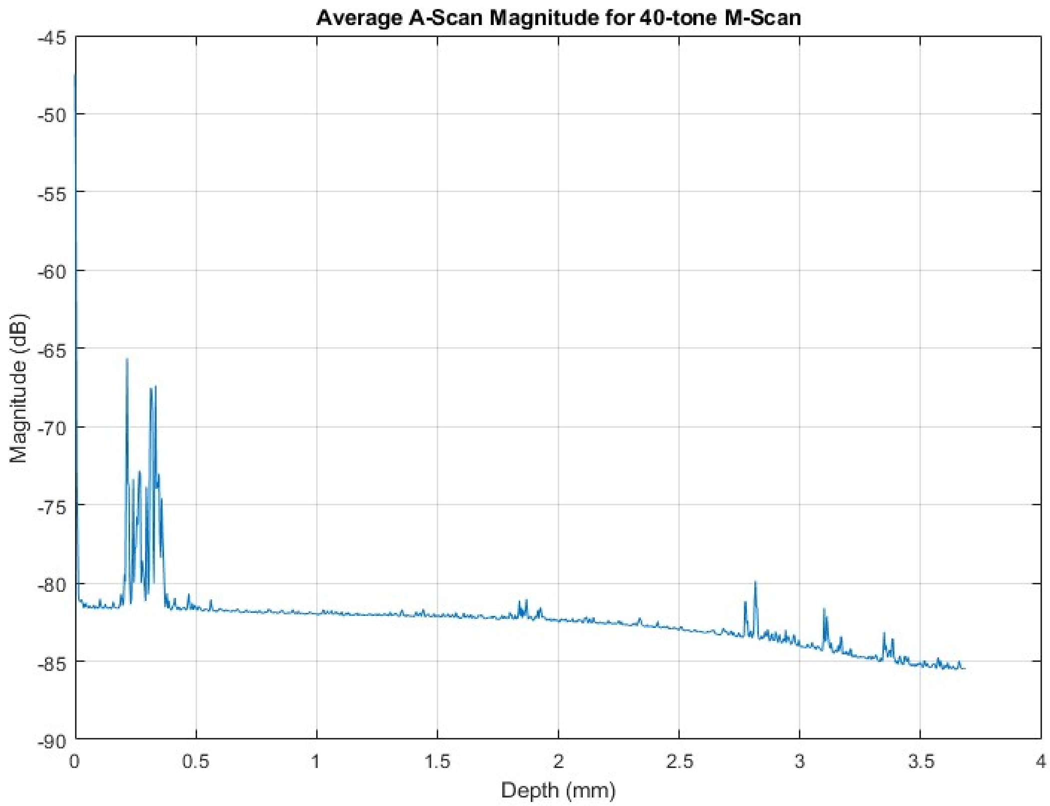


Figure 9: Average of the A-Scan magnitudes from the 40-tone M-Scan

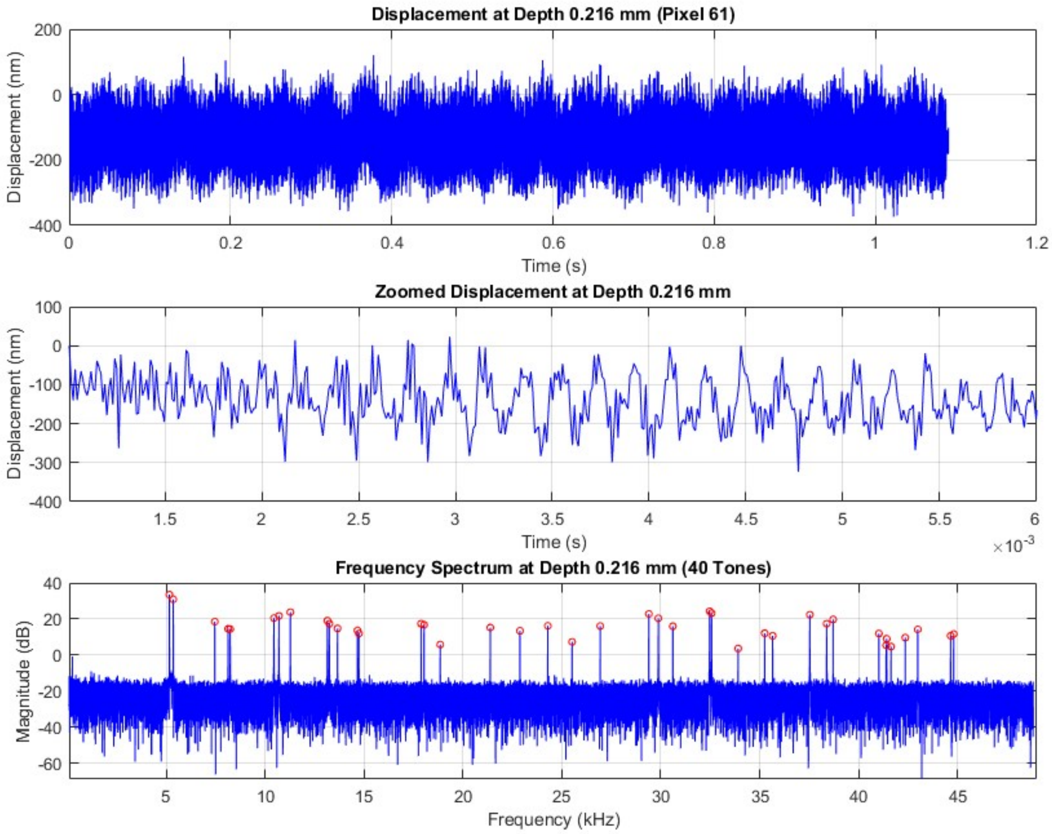


Figure 10: SDPM Time and Frequency Domain Data at Pixel 61 for 40-tone M-Scan

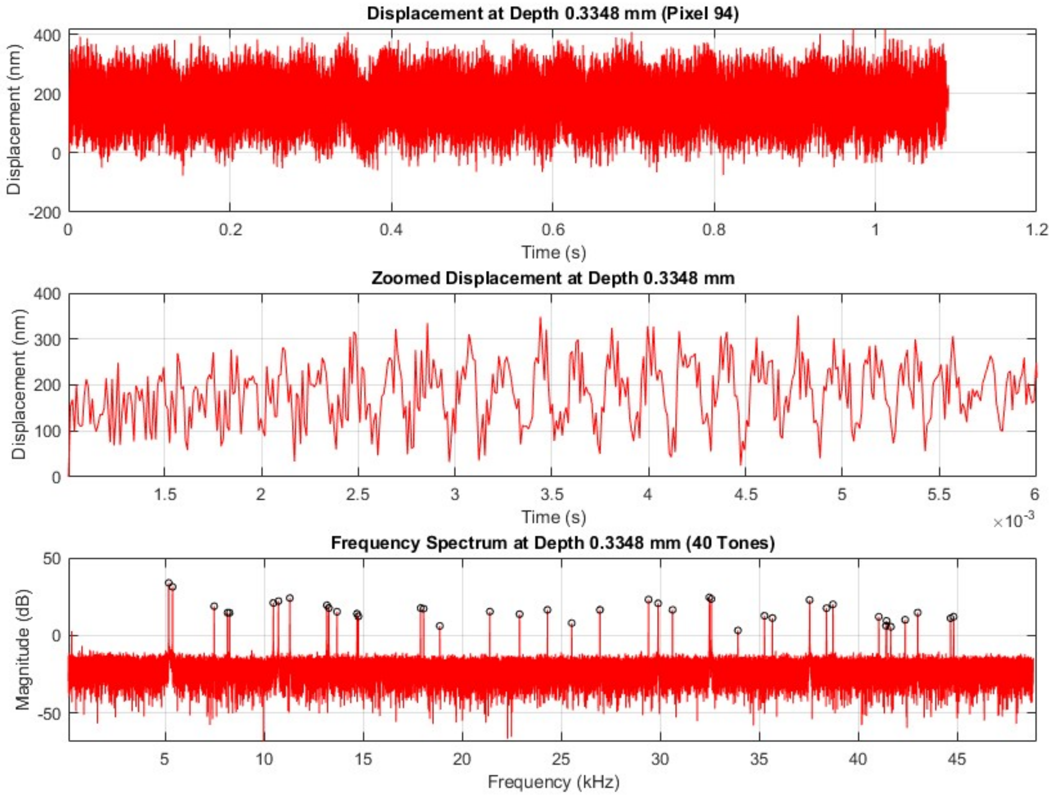


Figure 11: SDPM Time and Frequency Domain Data at Pixel 94 for 40-tone M-Scan

In the time domain, the displacement appears as a dense interference pattern which is expected as a result of a superposition of 40 different tones. This is further supported by the zoomed in view where the jaggedness reflects the summation of 40 different tones where the lower frequencies contribute to the broader structure, and the higher frequencies contribute to the sharp fluctuations. In the frequency spectrum, we observe 40 distinct peaks (circled in red and black) corresponding to the 40 different tones. Using MATLAB, the tones are displayed in Table 2:

Table 2: Sorted Tone Frequencies (kHz)

| **Tone** | **Freq. (kHz)** | **Tone** | **Freq. (kHz)** | **Tone** | **Freq. (kHz)** | **Tone** | **Freq. (kHz)** |
|---|---|---|---|---|---|---|---|
| 1 | 5.1619 | 11 | 13.6620 | 21 | 26.9404 | 31 | 38.3850 |
| 2 | 5.3574 | 12 | 14.6689 | 22 | 29.3966 | 32 | 38.7108 |
| 3 | 7.4565 | 13 | 14.7405 | 23 | 29.8775 | 33 | 41.0228 |
| 4 | 8.1321 | 14 | 17.8841 | 24 | 30.6090 | 34 | 41.3909 |
| 5 | 8.2385 | 15 | 18.0346 | 25 | 32.4676 | 35 | 41.4175 |
| 6 | 10.4478 | 16 | 18.8524 | 26 | 32.5668 | 36 | 41.6369 |
| 7 | 10.7002 | 17 | 21.3829 | 27 | 33.9114 | 37 | 42.3583 |
| 8 | 11.2775 | 18 | 22.8808 | 28 | 35.2514 | 38 | 42.9888 |
| 9 | 13.1508 | 19 | 24.2888 | 29 | 35.6516 | 39 | 44.6501 |
| 10 | 13.2481 | 20 | 25.5178 | 30 | 37.5323 | 40 | 44.8043 |

At different frequencies, the amplitudes differ. Observe in Figure 11 the tone at 5.1619kHZ which has an amplitude of roughly 33.89dB compared to the tone at 33.91kHZ which has an amplitude of 3.29dB. This signifies that the speaker is not all-pass otherwise the amplitude across all the tones would be the same.

The speaker is approximately linear at this nanometer scale. This is because we can clearly distinguish the 40 tones in the frequency spectrum. If the speaker were not linear, the 40 tones would not necessarily be the only spectral components present.

## Conclusion

This project demonstrates how SD-OCT data can be processed into meaningful A-scan, B-scan, and M-scan results. Using the developed pipeline, layered Scotch tape was imaged and microscopic speaker vibrations were tracked. From the B-scan, 10 tape layers were clearly resolved, and from the M-scan, both a single 8.01 kHz tone and a more complex 40-tone input were successfully recovered. Overall, these results highlight the effectiveness and sensitivity of SD-OCT for both structural imaging and motion analysis.